

Nassau Candy

Distributor

Factory-to-Customer Shipping Route Efficiency Analysis

Research Report & Exploratory Data Analysis

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10,194	1,320d	33.1%	\$93.4K
Total Orders	Avg Lead Time	Delay Rate	Gross Profit

This report presents a full exploratory data analysis, route efficiency benchmarking, geographic bottleneck identification, and shipping mode comparison for Nassau Candy Distributor's national logistics operations.

1. Background & Project Overview

Nassau Candy Distributor operates as a national distributor, shipping confectionery products from five specialized factories to customers across all major US regions. The company manages three product divisions — Chocolate, Sugar, and Other — spanning 15 distinct product lines. Despite rich transactional data, logistics decisions were historically made without route-level efficiency intelligence, creating a reactive rather than data-driven operational environment.

1.1 Problem Statement

The organization lacked clarity on:

- Which factory-to-customer routes consistently deliver on time
- Which routes experience frequent shipping delays
- How performance varies by region, state, and shipping method
- Where operational bottlenecks exist geographically
- Whether premium shipping modes (First Class, Same Day) actually outperform standard options

1.2 Dataset Overview

Field	Description	Type
Order ID	Unique order identifier	String
Order Date	Date order was placed	Date
Ship Date	Date order was shipped	Date
Ship Mode	Standard / First / Second / Same	Categorical
State/Province	Customer delivery state	Categorical
Region	Atlantic / Gulf / Interior / Pacific	Categorical
Product Name	One of 15 Wonka product lines	Categorical
Division	Chocolate / Sugar / Other	Categorical
Sales	Revenue per order (\$)	Numeric
Gross Profit	Revenue minus cost (\$)	Numeric
Cost	Manufacturing cost (\$)	Numeric
Units	Quantity per order	Numeric

2. Data Cleaning & Validation

A thorough data quality assessment was conducted prior to analysis. The dataset proved exceptionally clean with no missing values across all 18 columns and 10,194 records.

Check	Result	Action Taken
Missing Values	0 across all 18 columns	None required
Duplicate Records	0 duplicates	None required
Negative Lead Times	0 records	None required
Date Format	DD-MM-YYYY (dayfirst=True)	Parsed correctly
Negative Sales	0 records	None required
Invalid Product Names	All 15 products mapped to factory	Factory col created

Date fields were parsed with dayfirst=True to correctly interpret the DD-MM-YYYY format. All numerical fields (Sales, Gross Profit, Cost, Units) contained only non-negative values. Product names were validated against the factory-product correlation table provided in the project specification, confirming 100% coverage.

3. Feature Engineering

Three primary derived features were engineered to support the route efficiency analysis:

Feature	Formula / Method	Purpose
Lead Time (days)	Ship Date - Order Date	Primary KPI for efficiency
Factory	Product-to-factory lookup table	Route definition
Route	Factory + " to " + State/Province	Route-level aggregation
Route_Region	Factory + " to " + Region	Regional aggregation
Delayed (bool)	Lead Time > Mean + 1 Std Dev (1,583 days)	Delay flag for filtering
Order Month	Order Date period (YYYY-MM)	Time-series analysis

The delay threshold was set at Mean + 1 Standard Deviation (1,583.3 days), which statistically identifies shipments in the upper tail of the lead time distribution. This yielded a delay rate of 33.1%, consistent with the operational concern flagged in the project brief.

4. Exploratory Data Analysis (EDA)

4.1 Lead Time Overview

Lead times range from 904 to 1,642 days, with a mean of 1,320.8 days and standard deviation of 262.4 days. The distribution is bimodal — clustering around the 900-950 day band (fast routes) and the 1,600-1,640 day band (slow routes) — suggesting two fundamentally different operational pipelines exist within the company.

Statistic	Value
Minimum Lead Time	904 days
25th Percentile	1,271 days
Median	1,274 days
Mean	1,320.8 days
75th Percentile	1,638 days
Maximum Lead Time	1,642 days
Std Deviation	262.4 days
Delay Threshold	1,583.3 days (Mean + 1 Std)

4.2 Sales & Profitability

Total revenue across the analysis period (Jan 2024 – Dec 2025) was \$141,783.63 with a gross profit of \$93,442.80, reflecting a healthy 65.9% gross margin. The Chocolate division generates the highest order volume, driven by Lot's O' Nuts and Wicked Choccy's factories.

Metric	Value
Total Revenue	\$141,783.63
Total Gross Profit	\$93,442.80
Gross Margin	65.9%
Total Orders	10,194
Avg Order Value	\$13.91
Avg Profit/Order	\$9.17

4.3 Order Distribution by Segment

Dimension	Segment	Orders	Share
Region	Interior	3,583	35.1%
Region	Atlantic	3,178	31.2%
Region	Pacific	2,327	22.8%
Region	Gulf	1,106	10.8%
Ship Mode	Standard Class	5,943	58.3%
Ship Mode	Second Class	1,908	18.7%
Ship Mode	First Class	1,524	14.9%
Ship Mode	Same Day	819	8.0%
Division	Chocolate	8,397	82.4%
Division	Sugar	1,530	15.0%
Division	Other	267	2.6%

5. Route Aggregation & Benchmarking

5.1 Top 10 Most Efficient Routes

Routes are ranked by average lead time. The most efficient routes are predominantly served by Secret Factory and The Other Factory, both of which have smaller order volumes but significantly faster delivery pipelines.

Route	Avg Lead (days)	Shipments	Delay Rate	Eff. Score
Secret Factory → Nebraska	906.0	1	0.0%	100.0
Secret Factory → New Mexico	906.0	1	0.0%	100.0
The Other Factory → Louisiana	907.0	2	0.0%	99.9
The Other Factory → Connecticut	907.5	2	0.0%	99.9
Secret Factory → Mississippi	908.0	1	0.0%	99.8
Secret Factory → South Carolina	909.0	2	0.0%	99.8
Lot's O' Nuts → Illinois	909.0	47	0.0%	99.8
Wicked Choccy's → Kentucky	909.0	31	0.0%	99.8
Secret Factory → Iowa	909.5	2	0.0%	99.8
The Other Factory → Indiana	910.0	3	0.0%	99.7

5.2 Bottom 10 Least Efficient Routes

Route	Avg Lead (days)	Shipments	Delay Rate	Eff. Score
Sugar Shack → New Jersey	1642.0	1	100%	0.0
Secret Factory → New Hampshire	1641.0	1	100%	0.1
Sugar Shack → Connecticut	1641.0	1	100%	0.1
Wicked Choccy's → West Virginia	1639.0	3	100%	0.4
Lot's O' Nuts → North Dakota	1638.2	5	100%	0.5
Lot's O' Nuts → Maine	1638.0	3	100%	0.5
Lot's O' Nuts → Rhode Island	1638.0	2	100%	0.5
Wicked Choccy's → Alaska	1637.5	2	100%	0.6
Lot's O' Nuts → Mississippi	1637.0	5	100%	0.7

Route	Avg Lead (days)	Shipments	Delay Rate	Eff. Score
Secret Factory → Oklahoma	1636.0	2	100%	0.8

6. Geographic Bottleneck Analysis

Geographic analysis reveals that shipping inefficiency is not uniformly distributed across US regions. The Pacific and Interior regions consistently underperform, despite the Interior region hosting the highest order volume.

Region	Avg Lead Time	Order Volume	Delay Rate	Total Sales
Atlantic	1,322.7 days	3,178	32.3%	\$44,532
Gulf	1,311.4 days	1,106	32.0%	\$15,218
Interior	1,323.1 days	3,583	33.6%	\$50,198
Pacific	1,322.2 days	2,327	34.2%	\$31,836

The Gulf region has the lowest average lead time (1,311.4 days) and second-lowest delay rate. The Pacific region has the highest delay rate (34.2%) despite moderate order volume. This suggests geographic distance from factories is not the only driver — carrier performance and route-specific logistics play a significant role.

6.1 High-Risk States (Bottlenecks)

States approaching the maximum lead time ceiling represent critical bottlenecks. These require immediate logistics review:

- New Jersey — Routes from Sugar Shack hitting 1,642 day maximum
- Connecticut — Both Sugar Shack and The Other Factory routes delayed
- New Hampshire — Secret Factory route at 1,641 days (100% delay rate)
- West Virginia — Wicked Choccy's routes consistently slow
- North Dakota, Maine, Rhode Island — Lot's O' Nuts routes bottlenecked

6.2 Factory Performance by Geography

Factory	Avg Lead Time	Order Volume	Delay Rate	Best Region
Lot's O' Nuts	1,321.2 days	5,692	33.2%	Gulf
Wicked Choccy's	1,321.1 days	4,152	33.0%	Gulf
Secret Factory	1,321.9 days	217	31.8%	Interior
The Other Factory	1,280.3 days	100	26.0%	Gulf
Sugar Shack	1,340.0 days	33	48.5%	Atlantic

7. Ship Mode Performance Analysis

A critical counter-intuitive finding emerges from ship mode comparison: Standard Class delivers the fastest average lead times, while First Class and Same Day actually perform worse. This anomaly suggests that premium shipping modes may not be receiving prioritized handling in the current logistics setup.

Ship Mode	Avg Lead Time	Median Lead	Std Dev	Volume	Delay Rate
Standard Class	1,314.3 days	1,273 days	263.7	5,943	32.4%
Second Class	1,323.8 days	1,274 days	264.0	1,908	33.5%
Same Day	1,333.4 days	1,275 days	264.1	819	34.1%
First Class	1,338.3 days	1,275 days	263.5	1,524	33.9%

Standard Class outperforms all premium modes by 19–24 days on average. The similarity in median lead times (~1,273-1,275 days) across all modes but divergence in means suggests that premium modes occasionally experience extreme delays that inflate their averages. The cost-benefit case for upgrading to First Class or Same Day cannot currently be justified.

8. Key Performance Indicators Summary

KPI	Formula	Value	Target
Avg Shipping Lead Time	Mean(Ship Date - Order Date)	1,320.8 days	< 1,300 days
Delay Rate	% orders > 1,583 days lead time	33.1%	< 20%
Route Efficiency Score	Normalized lead-time index (0-100)	0 - 100	> 70 avg
On-Time Delivery Rate	1 - Delay Rate	66.9%	> 80%
Best Factory Lead Time	Min factory avg lead time	1,280.3 days	< 1,300 days
Worst Factory Lead Time	Max factory avg lead time	1,340.0 days	< 1,350 days
Top Region Delay Rate	Max regional delay rate	34.2% (Pacific)	< 25%

9. Key Insights & Recommendations

9.1 Critical Findings

Standard Class outperforms premium ship modes

Standard Class avg lead time is 1,314 days vs 1,338 for First Class — a 24-day gap. Operational priority assignment likely favors high-volume standard shipments.

The Other Factory is the efficiency leader

With only 100 orders, The Other Factory achieves an avg of 1,280 days — 40 days faster than its peers. Its logistics approach should be studied and replicated.

Sugar Shack is a high-risk supplier

Despite the lowest order volume (33 orders), Sugar Shack has the worst avg lead time (1,340 days) and highest delay rate (48.5%). This creates a supply chain vulnerability.

Pacific region is the most congested

At 34.2% delay rate and 1,322 avg days, the Pacific region consistently underperforms despite not being the highest-volume region.

Bimodal lead time distribution signals two operating modes

The clear bimodal distribution (peaks at ~910 days and ~1,635 days) suggests two distinct logistical pipelines are in operation, with no gradient in between.

9.2 Strategic Recommendations

- Audit The Other Factory logistics model — identify what makes it faster and apply to all factories.
- Conduct urgent review of Sugar Shack supply chain — consider capacity expansion or sourcing backup.
- Establish regional distribution hubs in Pacific and Interior to reduce last-mile lead times.
- Reassign premium ship mode (First Class, Same Day) shipments to expedited carrier contracts.
- Flag 14 critical bottleneck routes (all above 1,635 days) for immediate carrier renegotiation.
- Implement real-time delay prediction — the Streamlit dashboard enables proactive monitoring.
- Target the 3,379 currently delayed orders for customer communication and retention follow-up.

This analysis transforms raw order data into actionable logistics intelligence. Implementing these recommendations is projected to reduce the delay rate from 33.1% toward the 20% industry benchmark, improving customer satisfaction and reducing operational cost across all four US regions.